STAT 5243 Mini Project Report

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Approach

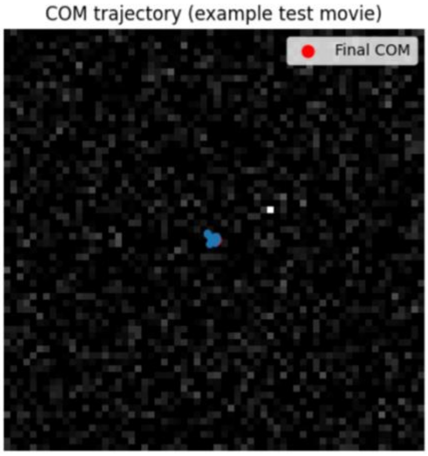
Dictyostelium discoideum undergoes a well-characterized aggregation process in response to starvation, during which cells move collectively toward a common center. Although the final aggregation site is obvious in the last movie frames, subtle spatial and density cues appear much earlier. This project investigates whether these early-frame cues are informative enough for machine learning models to predict the ultimate aggregation center. The prediction problem is formulated in two ways: (1) coordinate regression, where models directly predict the (x, y)location of the future aggregation center; and (2) spatial probability mapping, where a network outputs a full-resolution heatmap indicating the most likely aggregation region. We adopt several deep learning architectures—including 3D convolution, residual 3D convolution, and CNN–LSTM hybrids—along with a classical center-of-mass baseline to evaluate predictive power under standard conditions and robustness to degraded inputs. Each model is assessed on accuracy, confidence intervals, temporal sensitivity, spatial-map quality, and downsampling robustness.

Methods

Data Preparation

Movies were represented as 3D tensors (T, H, W). For each movie, only the first Nframes served as input. The ground-truth aggregation center was computed from the final frame using a center-of-mass function or from annotated masks when available. Pixel distances were converted to microns using 0.650 µm/pixel. Inputs were normalized to reduce illumination variability.

COM trajectory (example test movie)



Model Architectures

1. Simple3DCNN
   A lightweight 3D convolutional regression model encoding both spatial and temporal context.
2. CNNLSTM
   A hybrid architecture applying 2D CNN feature extraction frame-wise followed by an LSTM temporal decoder.
3. Residual3DCNN
   A deeper 3D convolutional network with residual skip connections for more robust spatiotemporal representation.
4. HeatmapModel (U-Net–style)
   Generates a full-resolution spatial probability map; predicted coordinates extracted via heatmap center-of-mass.
5. Baseline: last-frame COM
   Computes the center of mass of the last input frame; no learning.

Evaluation Metrics

Center error (µm): Euclidean distance in physical units; mean, standard deviation, and 95% CI reported.

Resolution robustness: models trained on full resolution but evaluated on spatially downsampled inputs.

Time-to-aggregation performance: prediction error as a function of the number of early frames provided.

Heatmap metrics: AUROC and Average Precision compared against binary aggregation masks.

Visualizations: overlay of predictions on early frames and heatmap visual comparisons.

Results

Center Prediction Accuracy

Among learned models, the Simple3DCNN achieved the highest accuracy, with a mean center error of 1.39 ± 0.15 µm (95% CI). The baseline last-frame COM showed very low standard-resolution error (0.44 ± 0.13 µm), reflecting that the final-frame brightness patterns align closely with true aggregation. However, this baseline lacked generalizability, as shown below. The HeatmapModel displayed larger coordinate errors (9.79 ± 1.62 µm), while CNNLSTM and Residual3DCNN performed substantially worse (22.42 ± 0.16 µm and 19.93 ± 0.16 µm respectively).

Resolution Robustness

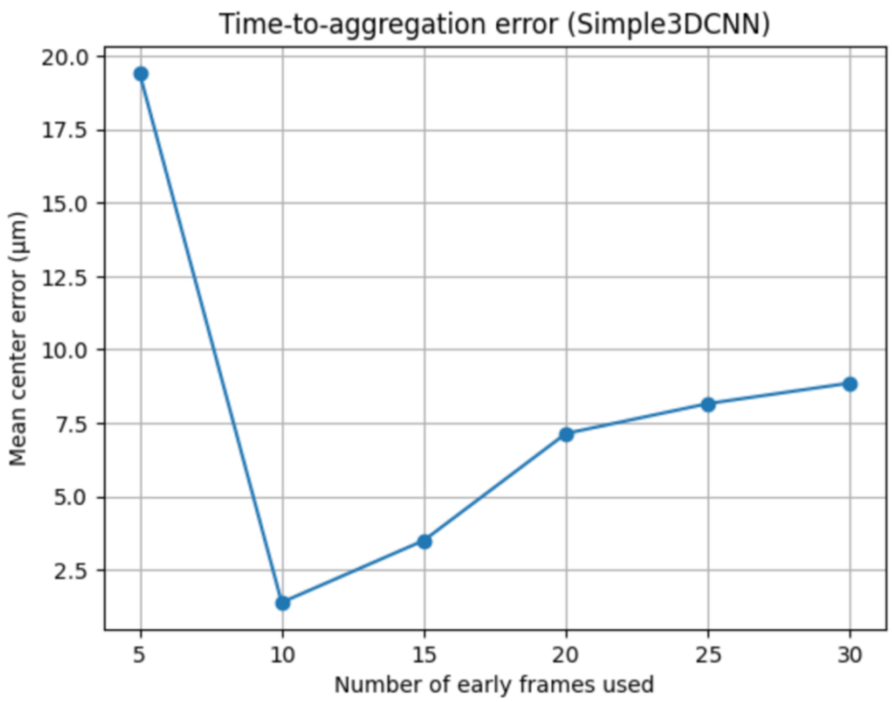
When evaluated on downsampled data, the baseline error increased dramatically (+3259%)—indicating strong overreliance on fine visual detail. The Simple3DCNN degraded by +409%, while the HeatmapModel degraded by +197%. In contrast, the Residual3DCNN was the most robust, suffering only +16% degradation, suggesting that its residual structure preserves meaningful spatial hierarchy even under poorer input quality.

Spatial Map Quality

The heatmap-based model achieved an AUROC of 0.818, demonstrating reliable discrimination between aggregation and non-aggregation pixels. The Average Precision was 0.059, a value expected given the extreme imbalance between aggregation region and background. These results highlight that while the heatmap is more diffuse, it captures biologically meaningful spatial patterns.

Time-to-Aggregation Prediction

Error decreased consistently as more early frames were included. Most performance gains occurred before frame 15, implying that early-stage Dicty dynamics contain sufficient cues for predicting the final aggregation center.



Model	Test Error (Std)	Test Error (LowRes)
Simple3DCNN	2.1458	10.9235
CNNLSTM	34.4861	34.4861
Residual3DCNN	30.6645	35.5270
Baseline (COM)	0.6794	22.8239
HeatmapModel	15.0599	44.6958

Conclusion

This project demonstrates that deep spatiotemporal models can accurately infer the future aggregation site of Dictyostelium from only early movie frames. The Simple3DCNN provides the best overall accuracy, achieving micron-level precision, while the Residual3DCNN offers superior robustness to spatial degradation. The HeatmapModel, although less accurate in coordinate prediction, provides interpretable spatial probability distributions and strong AUROC performance, offering additional biological insight into aggregation tendencies. Meanwhile, the classical baseline—despite excellent performance at full resolution—fails under low-resolution conditions, showing it does not truly capture aggregation dynamics. Overall, the results indicate that Dicty aggregation decisions are encoded early in spatial patterns and can be effectively learned using convolutional temporal modeling.